

Notes: Liu, Stambaugh, and Yuan (JFE, 2019)

Size and Value in China

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1 Big picture

- **Question.** What are the right size and value factors for China, and can they explain Chinese stock-return anomalies better than a direct replication of the Fama–French U.S. procedure?
- **Core point.** China is not simply a smaller version of the U.S. equity market. Its IPO regulation, reverse-merger market, retail investor base, short-sale constraints, and accounting/information environment change how factor construction should work.
- **Size result.** The smallest 30% of listed firms in China are contaminated by potential shell value. These firms are attractive reverse-merger shells because China’s IPO approval process is tight. Their stock prices therefore reflect listing-option value in addition to operating business value.
- **Value result.** The earnings-price ratio, not book-to-market, is the dominant value characteristic in China. In Fama–MacBeth horse races, positive earnings yield subsumes BM, AM, and CP.
- **Factor model.** The paper proposes a China three-factor model, CH-3, with a market factor, a size factor that excludes the smallest 30% of stocks, and an EP-based value factor.
- **Benchmark comparison.** CH-3 strongly dominates a three-factor model made by mechanically replicating Fama and French (1993) in China. The FF-style model cannot price CH-3’s size and value factors; CH-3 can price the FF-style factors.
- **Anomaly result.** CH-3 explains most previously documented Chinese anomalies, including profitability and volatility anomalies that are difficult for U.S. three-factor models.
- **Extension.** A four-factor model, CH-4, adds a turnover-based sentiment factor. This factor helps explain reversal and turnover anomalies that CH-3 leaves partially unexplained.
- **Takeaway.** Factor models should be built around the institutions and economic mechanisms of the market being studied. In China, shell value and earnings yield are central.

2 Incremental contribution

2.1 Relative to Fama–French style factor construction

- Earlier Chinese asset-pricing studies often imported the Fama–French three-factor construction almost mechanically, using BM for value and including the entire cross-section in size sorts.
- This paper shows that direct importation fails because China’s market has a large shell-value component among very small stocks and because EP, not BM, captures the strongest value dimension.
- The paper therefore contributes a market-specific factor construction rather than a direct application of U.S. empirical recipes.

Interpretation: The contribution is methodological and economic: factor construction is not only a sorting procedure; it embeds assumptions about what firm characteristics capture systematic risk or mispricing. In China, those assumptions differ from the U.S. setting.

2.2 Relative to the Chinese anomaly literature

- The paper consolidates evidence on many anomalies previously reported in China: size, value, profitability, volatility, reversals, turnover, investment, accruals, and illiquidity.
- It re-tests these anomalies under a unified post-2000 sample and data source, then asks whether a factor model can explain their long-short returns.

- CH-3 explains eight of ten significant CAPM anomalies, while the mechanically replicated FF-3 explains only three.
- CH-4 further addresses turnover and reversal anomalies by adding a turnover-based sentiment factor.

Interpretation: The incremental contribution is not just proposing new factors. It provides a benchmark model that reorganizes the Chinese anomaly literature and shows which anomalies remain after controlling for China-specific size and value factors.

2.3 Relative to research on Chinese institutions

- The paper links asset pricing to China’s IPO approval regime and reverse-merger market.
- It documents that reverse-merger shells are overwhelmingly drawn from the smallest listed firms.
- It estimates that shell value can represent roughly 30% of market value for typical bottom-30% firms.
- It then shows that including these firms distorts both size and value premiums.

Interpretation: This is an institutional contribution. The paper identifies a specific regulatory mechanism that contaminates an apparently standard characteristic, firm size, and then modifies factor construction accordingly.

3 Data and measurement

3.1 Chinese equity universe

- The core sample is China’s A-share market from January 2000 through December 2016.
- A-shares are domestically traded stocks and are the relevant market for Chinese domestic investors.
- The paper uses CSMAR and WIND-type Chinese market data sources for returns, accounting variables, market values, IPO activity, and reverse-merger transactions.
- The market factor is based on the top 70% of listed stocks by market capitalization, after excluding the smallest 30%.

Interpretation: The paper starts in 2000 because Chinese accounting and return data are more reliable in the post-2000 period, and because China-specific anomalies are mostly documented over modern samples.

3.2 Shell value and reverse mergers

- China’s IPO process is highly regulated and historically slow.
- Private firms can bypass IPO queues through reverse mergers: they gain control of an existing listed shell, inject assets into it, and thereby obtain a listing.
- Listed firms with small market capitalizations are attractive shells because they are cheaper to acquire.
- The paper estimates potential shell value by combining the probability of becoming a reverse-merger target with the average price appreciation from that event.

Interpretation: A small Chinese listed firm is not just a small operating company. It can also be an option on a scarce listing status. This makes the bottom tail of size special.

3.3 Size measurement

The paper uses total market capitalization when sorting on size:

$$ME_{i,t} = P_{i,t} \times SharesOutstanding_{i,t}. \quad (1)$$

The key construction step is:

$$Universe_t^{CH} = \{i : ME_{i,t} \text{ is above the 30th percentile}\}. \quad (2)$$

Interpretation: Excluding the smallest 30% removes the segment most contaminated by shell value while retaining about 93% of aggregate market capitalization.

3.4 Value ratios

The paper compares four valuation ratios:

- book-to-market, BM ;
- assets-to-market, AM ;
- earnings-price, EP ;
- cash-flow-to-price, CP .

For ratios that can be negative, the paper uses positive components and negative-value indicators:

$$EP^+ = \max(EP, 0), \quad D(EP < 0) = \mathbf{1}\{EP < 0\}. \quad (3)$$

Interpretation: EP is the key value measure. In Chinese cross-sectional return regressions, EP dominates BM, AM, and CP.

4 Factor construction and empirical specifications

4.1 CH-3 factor construction

Each month, after excluding the smallest 30% of stocks:

- stocks are split into two size groups: small S and big B ;
- stocks are split into three EP groups: value V , middle M , and growth G ;
- six value-weighted portfolios are formed from the intersections: S/V , S/M , S/G , B/V , B/M , and B/G .

The size factor is:

$$SMB = \frac{1}{3}(S/V + S/M + S/G) - \frac{1}{3}(B/V + B/M + B/G). \quad (4)$$

The value factor is:

$$VMG = \frac{1}{2}(S/V + B/V) - \frac{1}{2}(S/G + B/G). \quad (5)$$

The market factor is:

$$MKT = R_{top70}^{vw} - R^f, \quad (6)$$

where R^f is the one-year deposit interest rate.

Interpretation: The CH-3 model is a China-specific analogue of FF-3, but it differs in two crucial dimensions: it removes shell-contaminated stocks and uses EP rather than BM.

4.2 FF-3 replication benchmark

The paper constructs a benchmark FF-3 model by applying the Fama–French procedure in China:

- use BM rather than EP to form the value factor;
- construct a size factor using the standard FF-style six portfolios;
- compare this benchmark against CH-3 in pricing factor returns and anomaly portfolios.

Interpretation: FF-3 is not presented as the authors’ preferred model. It is the natural benchmark because much of the prior Chinese literature used a mechanically replicated Fama–French model.

4.3 Earnings-surprise regression

To show that the smallest stocks are less connected to operating fundamentals, the paper estimates:

$$R_{i,t-k,t+k} = a + bSUE_{i,t} + e_{i,t}, \quad (7)$$

where $R_{i,t-k,t+k}$ is the event-window return around an earnings announcement and SUE is standardized unexpected earnings.

Interpretation: If very small stocks are priced partly as listing shells, their returns should respond less to earnings surprises. Table 1 confirms that earnings surprise explains less of their returns than for larger stocks.

4.4 Fama–MacBeth valuation-ratio regressions

The value-ratio horse race takes the form:

$$R_{i,t+1} = a_t + b_t\beta_{i,t} + c_t \log ME_{i,t} + d_t X_{i,t} + e_{i,t+1}, \quad (8)$$

where X includes combinations of $\log BM$, $\log AM$, EP^+ , $D(EP < 0)$, CP^+ , and $D(CP < 0)$.

Interpretation: EP remains strong when other ratios are included, while the other value ratios lose explanatory power. This motivates the EP-based value factor.

4.5 Factor-model alpha tests

For a test asset or factor R_t , the paper estimates:

$$R_t = \alpha + \beta_{MKT}MKT_t + \beta_{SMB}SMB_t + \beta_{VMG}VMG_t + \epsilon_t. \quad (9)$$

The benchmark FF-3 regression replaces SMB and VMG with $FFSMB$ and $FFHML$.

Interpretation: A factor model prices a test asset when the alpha is economically small and statistically insignificant. CH-3 prices the FF-style factors; FF-3 fails to price the CH-3 factors.

4.6 Anomaly tests

For each anomaly, stocks are sorted into deciles and the long-short portfolio return is formed as:

$$R_t^{anom} = R_t^{long} - R_t^{short}. \quad (10)$$

The paper estimates CAPM, CH-3, FF-3, and CH-4 alphas for these long-short spreads.

Interpretation: The anomaly tests ask whether the factor model can explain patterns not used directly in factor construction.

4.7 CH-4 turnover factor

The CH-4 model adds a sentiment-motivated turnover factor:

$$R_t^{anom} = \alpha + \beta_{MKT}MKT_t + \beta_{SMB}SMB_t + \beta_{VMG}VMG_t + \beta_{PMO}PMO_t + \epsilon_t. \quad (11)$$

PMO is “pessimistic minus optimistic,” based on abnormal turnover.

Interpretation: In China, high abnormal turnover is interpreted as a proxy for speculative optimism under short-sale constraints. The factor helps absorb turnover and reversal anomalies.

5 Identification, interpretation, and limits

5.1 Why the shell mechanism matters for identification

- The shell mechanism gives an institutional reason why the smallest stocks should behave differently.
- The reverse-merger evidence shows that shell targets are concentrated in the bottom size deciles.
- The earnings-surprise evidence shows that returns of the smallest stocks are less linked to operating fundamentals.
- The IPO activity evidence links shell values to IPO constraints.

Interpretation: The size adjustment is not arbitrary data mining. It is tied to a market institution: scarce listings created by IPO regulation.

5.2 Why EP matters for interpretation

- In the U.S., BM emerged as the preferred value variable in Fama and French (1992).
- In China, the horse race points instead to EP.
- EP also captures profitability-like and value-like effects, explaining why CH-3 can handle profitability anomalies.

Interpretation: This is why the paper’s value factor is not merely an alternative sort. It changes the economic content of the factor.

5.3 What the paper does not prove

- The paper does not prove whether factor premia are risk compensation or mispricing.
- The shell-value estimate relies on reverse-merger frequencies and price reactions.
- Factor performance is evaluated in-sample over 2000–2016.
- China’s market institutions have evolved over time, so factor definitions may need updating if IPO constraints or reverse-merger rules change.

Interpretation: The paper’s contribution is a strong empirical factor model for China, not a final theory of why those factors command premia.

6 Tables and figures

6.1 Figure 1 and Figure 2: shell value and the smallest firms

Read:

- Figure 1 shows that reverse-merger shells are overwhelmingly small firms; about half come from the smallest size decile and 83% from the smallest 30%.
- Figure 2 shows that estimated shell value is economically large and time-varying.
- Panel A of Figure 2 shows the ratio of shell value to market capitalization, often around 20–40% and sometimes higher.
- Panel B shows the estimated RMB value of listing status, rising strongly after 2014.

Economic message: The smallest firms are not clean size stocks. Their valuations contain a large component associated with being listed shells.

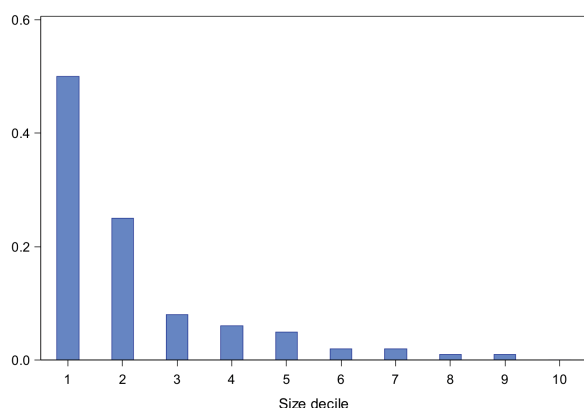


Figure 1: size distribution of reverse-merger shells

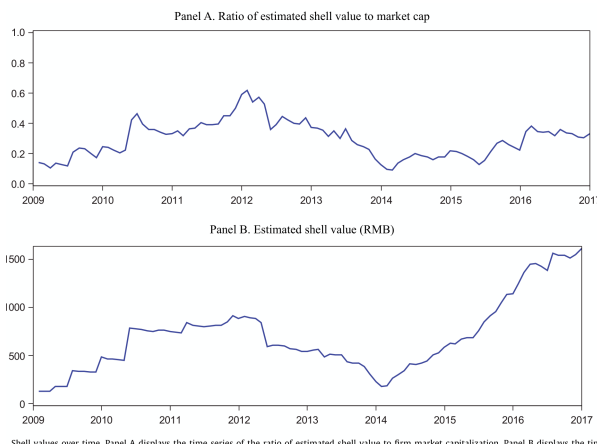


Figure 2: shell values over time

6.2 Table 1 and Table 2: fundamentals and value ratios

Read:

- Table 1 shows that earnings-surprise coefficients and R^2 values are smallest for the smallest Chinese stocks.
- This supports the view that shell values weaken the link between stock returns and operating fundamentals.
- Table 2 shows that EP dominates BM, AM, and CP in Fama–MacBeth regressions.
- In the full regression, the coefficient on EP^+ remains strong, while BM, AM, and CP become much weaker.

Economic message: China-specific factor construction is supported by direct evidence: bottom-size stocks are contaminated by shell value and EP is the strongest value variable.

Table 1
Return reactions to earnings surprises across different size groups in China and the US.
The table reports slope estimates and R-squares in a panel regression of earnings-window returns on earnings surprises.

$R_{i,t-k,t+k} = a + bSUE_{i,t} + \epsilon_{i,t}$,
in which earnings are announced on day t ; $R_{i,t-k,t+k}$ is the cumulative return on stock i , in excess of the market return, over the surrounding trading days from $t-k$ through $t+k$; $SUE_{i,t} = \Delta_{i,t}/\sigma(\Delta_{i,t})$; $\Delta_{i,t}$ equals the year-over-year change in stock i 's quarterly earnings; and $\sigma(\Delta_{i,t})$ is the standard deviation of $\Delta_{i,t}$ for the last eight quarters. Panel A contains results for $k=0$; Panel B contains results for $k=3$. The regression is estimated within each of three size groups in both the China and US markets. The groups are formed based on the top 30%, middle 40%, and bottom 30% of the previous month's market capitalizations. The sample periods are January 2000 through December 2016 for China and January 1980 through December 2016 for the US. The US returns data are from the Center for Research in Security Prices (CRSP) and the earnings data are from Compustat. White (1980) heteroskedasticity-consistent t -statistics are reported in parentheses. The estimates of b are multiplied by 100.

Quantity	China			US		
	Smallest	Middle	Largest	Smallest	Middle	Largest
Panel A: $k=0$						
b	0.14 (6.34)	0.17 (12.42)	0.24 (17.28)	0.19 (7.51)	0.07 (6.99)	0.05 (9.90)
R^2	0.003	0.010	0.017	0.005	0.003	0.002
Panel B: $k=3$						
b	0.43 (9.74)	0.58 (17.91)	0.59 (17.60)	0.52 (7.84)	0.20 (6.03)	0.13 (10.68)
R^2	0.006	0.016	0.021	0.012	0.005	0.003

Table 1: return reactions to earnings surprises

Table 2
Fama-MacBeth regressions of stock returns on beta, size, and valuation ratios.
The table reports average slope coefficients from month-by-month Fama-MacBeth regressions. Individual stock returns are regressed cross-sectionally on stock characteristics as of the previous month. The columns correspond to different regression specifications, with nonempty rows indicating the included regressors. The regressors include preanking CAPM β , estimated using the past 12 months of daily returns with a five-lag *Simon* (1975) correction; the log of month-end market cap ($\log M$); the log of book-to-market ($\log BM$); the log of assets-to-market ($\log AM$); EP^+ , which equals the positive values of earnings-to-price, and zero otherwise; $D(EP < 0)$, which equals one if earnings are negative, and zero otherwise; CP^+ , and $D(CP < 0)$ (with the last two similarly defined). The last row reports the average adjusted R-squared for each specification. The sample period is January 2000 through December 2016. The t -statistics based on Newey and West (1987) standard errors with four lags are reported in parentheses.

Quantity	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Intercept	0.0149 (1.94)	0.0581 (3.32)	0.0571 (3.19)	0.0659 (3.90)	0.0629 (3.74)	0.0690 (4.03)	0.0564 (3.19)	0.0716 (4.40)	0.0728 (4.39)
β	-0.0002 (-0.09)		-0.0010 (-0.27)	-0.0018 (-0.71)	-0.0017 (-0.67)	0.0002 (0.07)	-0.0010 (-0.37)	-0.0002 (-0.06)	-0.0004 (-0.15)
$\log ME$		-0.0049 (-2.91)	-0.0046 (-2.69)	-0.0046 (-2.73)	-0.0048 (-3.00)	-0.0068 (-4.34)	-0.0047 (-2.80)	-0.0066 (-4.49)	-0.0064 (-4.40)
$\log BM$				0.0057 (3.21)				0.0022 (1.31)	0.0035 (1.76)
$\log AM$					0.0045 (3.03)			0.0014 (0.99)	
EP^+						0.9503 (4.88)		0.7825 (4.38)	0.7960 (5.06)
$D(EP < 0)$						0.0006 (0.31)		-0.0005 (-0.29)	-0.0001 (-0.04)
CP^+							0.0546 (3.41)	0.0181 (1.35)	
$D(CP < 0)$							0.0019 (3.11)	0.0016 (2.37)	
R^2	0.0196	0.0277	0.0441	0.0652	0.0677	0.0615	0.0454	0.0832	0.0776

Table 2: Fama-MacBeth value-ratio regressions

6.3 Table 3 and Table 4: factor premia and explanatory power

Read:

- Table 3 reports monthly averages of 0.66% for MKT, 1.03% for SMB, and 1.14% for VMG.
- The size and value premiums are statistically significant, with t -statistics of 3.25 and 4.34.
- Table 4 shows that CH-3 explains 53.6% of the average individual-stock return variance in China, versus 38.5% for the market alone.
- In the U.S. comparison, FF-3 explains 27.3% versus 17.7% for the market alone.

Economic message: Size and value are quantitatively important in China. They carry large average premiums and add substantial explanatory power beyond the market factor.

Table 3

Summary statistics for the CH-3 factors.

This table reports the means, standard deviations, *t*-statistics, and pairwise correlations for the three factors in the CH-3 model. The means and standard deviations are expressed in percent per month. The sample period is January 2000 through December 2016 (204 months).

Factor	Mean	Std. dev.	<i>t</i> -stat.	Correlations		
				<i>MKT</i>	<i>SMB</i>	<i>VMG</i>
<i>MKT</i>	0.66	8.09	1.16	1.00	0.12	-0.27
<i>SMB</i>	1.03	4.52	3.25	0.12	1.00	-0.62
<i>VMG</i>	1.14	3.75	4.34	-0.27	-0.62	1.00

Table 3: CH-3 factor summary statistics

Table 4Average *R*-squares for individual stocks in China and the US.

The table compares the average *R*-squares in regressions of monthly individual stocks' returns on factors in China's and the US stock markets. Regressions are estimated for four models: one with just the excess market return (*MKT*); one with *MKT* plus the size factor; one with *MKT* plus the value factor; and the three-factor model with *MKT* plus the size and value factors. In China's stock market, we use our CH-3 model's market (*MKT*), size (*SMB*), and value (*VMG*) factors, while in the US market, we use FF-3's three factors: market, *SMB* and *BM*-based *HML*. For each stock, we run rolling-window regressions of each stock's monthly returns on factors over the past three years (36 months). We average the *R*-square across time for each stock and then compute the mean of these averages across all stocks. Panel A reports average *R*-squares across all individual stocks on China's main boards and the Growth Enterprise Market (GEM), including the bottom 30% of stocks. Panel B reports average *R*-squares of all but the smallest 30% of stocks. Panel C reports average *R*-squares of all common stocks from the NYSE, Amex, and Nasdaq for the US. The sample periods for both China and the US are from January 2000 through December 2016 (204 months).

Factors	Avg. <i>R</i> -square
<i>Panel A: All individual stocks in China</i>	
<i>MKT</i>	0.385
<i>MKT, SMB</i>	0.507
<i>MKT, VMG</i>	0.471
<i>MKT, SMB, VMG</i>	0.536
<i>Panel B: All but the smallest 30% of stocks in China</i>	
<i>MKT</i>	0.417
<i>MKT, SMB</i>	0.528
<i>MKT, VMG</i>	0.501
<i>MKT, SMB, VMG</i>	0.562
<i>Panel C: All individual stocks in the US</i>	
<i>MKT</i>	0.177
<i>MKT, SMB</i>	0.231
<i>MKT, HML</i>	0.226
<i>MKT, SMB, HML</i>	0.273

Table 4: return-variance explanatory power

6.4 Table 5 and Table 6: CH-3 versus FF-3 and CAPM anomalies

Read:

- Table 5 shows that FF-3 cannot price CH-3's size and value factors: CH-3 *SMB* and *VMG* have FF-3 alphas of 0.47% and 1.39% per month.
- CH-3 prices the FF-3 factors: *FFSMB* and *FFHML* have insignificant CH-3 alphas.
- Table 6 reports ten significant CAPM anomalies, including EP, BM, profitability, volatility, reversal, and turnover anomalies.
- The strongest CAPM alphas include EP, one-month return reversal, and abnormal turnover.

Economic message: The benchmark FF-style model misses the China-specific factors. The anomaly table establishes the set of return patterns that a useful Chinese factor model must explain.

Table 5

Abilities of models CH-3 and FF-3 to explain each other's size and value factors.

Panel A reports a factor's estimated monthly alpha (in percent) with respect to the other model (with [White, 1980](#), heteroskedasticity-consistent t -statistics in parentheses). Panel B computes the Gibbons–Ross–Shanken (1989) F -test of whether a given model produces zero alphas for the factors of the other model (p -value in parentheses). The sample period is January 2000 through December 2016.

Factors	Alphas with respect to:	
	CH-3	FF-3
Panel A: Alpha (t-statistic)		
FFSMB	-0.04 (-0.66)	-
FFHML	0.34 (0.97)	-
SMB	-	0.47 (7.03)
VMG	-	1.39 (7.93)
Panel B: GRS F-statistics (p-value)		
FFSMB, FFHML	0.88 (0.41)	-
SMB, VMG	-	33.90 (2.14×10^{-13})

Table 5: CH-3 versus FF-3 factor pricing

6.5 Table 7 and Table 8: CH-3 versus FF-3 anomaly pricing

Read:

- Table 7 shows that CH-3 leaves small and mostly insignificant alphas for value, profitability, and volatility anomalies.
- CH-3 struggles mainly with reversal and abnormal-turnover anomalies.
- Table 8 shows that FF-3 leaves large alphas across more anomaly categories, including EP, ROE, volatility, reversal, and turnover.
- FF-3 fails especially badly on the EP value anomaly because it uses BM rather than EP.

Economic message: CH-3 is not just better at pricing other factors; it also explains many economically important return anomalies in China.

Table 7
CH-3 alphas and factor loadings for anomalies.
For each of ten anomalies, the table reports the monthly long-short return spread's CH-3 alpha and factor loadings. For each anomaly, the regression estimated is

$$R_t = \alpha + \beta_{MKT} MKT_t + \beta_{SMB} SMB_t + \beta_{VMG} VMG_t + \epsilon_t,$$

where R_t is the anomaly's long-short return spread in month t , MKT_t is the excess market return, SMB_t is CH-3's size factor, and VMG_t is the EP-based value factor. In Panel A, for the unconditional sorts, the long leg of an anomaly is the value-weighted portfolio of stocks in the lowest decile of the anomaly measure, and the short leg contains the stocks in the highest decile, with a high value of the measure being associated with lower return. In Panel B, long/short legs are neutralized with respect to size. That is, we first form size deciles by sorting on the previous month's market value. Within each size decile, we then create ten deciles formed by sorting on the anomaly variable. Finally, we form the anomaly's decile portfolios, with each portfolio pooling the stocks in a given anomaly decile across the size groups, again with value weighting. Panel B omits the size anomaly, whose alpha equals zero by construction with size-neutral sorts. Our sample period is January 2000 through December 2016. All t -statistics are based on the heteroskedasticity-consistent standard errors of [White \(1980\)](#).

Category	Anomaly	α	β_{MKT}	β_{SMB}	β_{VMG}	$t(\alpha)$	$t(\beta_{MKT})$	$t(\beta_{SMB})$	$t(\beta_{VMG})$
Panel A: Unconditional sorts									
Size	Market cap	0.21	0.01	1.45	-0.54	1.71	0.77	40.88	-11.70
Value	EP	0.04	0.04	-0.38	1.40	0.16	1.31	-4.73	14.75
Value	BM	0.64	0.06	-0.03	0.43	1.02	0.65	-0.14	1.64
Value	CP	0.20	0.14	-0.28	0.64	0.45	2.10	-1.95	4.00
Profitability	ROE	-0.36	0.03	-0.29	1.28	-0.88	0.70	-2.35	9.43
Volatility	1-Month vol.	0.23	-0.23	-0.12	0.75	0.44	-3.81	-0.67	3.86
Volatility	MAX	0.27	-0.30	-0.05	0.48	0.65	-4.57	-0.30	2.55
Reversal	1-Month return	0.03	-0.06	0.56	0.01	1.70	-0.69	3.15	0.03
Turnover	12-Month turn.	0.42	-0.14	-0.85	0.77	1.30	-3.69	-9.33	7.90
Turnover	1-Mo. abn. turn.	1.28	-0.22	0.18	-0.16	2.86	-2.78	0.93	-0.76
Panel B: Size-neutral sorts									
Value	EP	0.23	0.02	0.05	1.32	0.82	0.47	0.57	11.97
Value	BM	0.61	0.13	-0.15	0.53	0.98	1.60	-0.80	2.19
Value	CP	0.18	0.11	-0.06	0.52	0.54	1.88	-0.50	3.84
Profitability	ROE	-0.37	0.05	0.41	1.20	-1.04	1.23	4.14	9.66
Volatility	1-Month vol.	0.20	-0.28	-0.08	0.64	0.42	-4.96	-0.49	3.34
Volatility	MAX	0.00	-0.26	0.05	0.45	0.00	-4.38	0.30	2.50
Reversal	1-Month return	1.13	0.01	0.41	0.10	2.12	0.11	2.59	0.55
Turnover	12-Month turn.	0.25	-0.22	-0.43	0.74	0.69	-4.94	-3.91	5.74
Turnover	1-Mo. abn. turn.	1.24	-0.18	0.25	-0.08	3.04	-2.79	1.55	-0.43

Table 7: CH-3 alphas and loadings for anomalies

Table 6

CAPM alphas and betas for anomalies.

For each of ten anomalies, the table reports the monthly long-short return spread's, average ($\bar{R}$), CAPM alpha (α), and CAPM beta (β). In Panel A, for the unconditional sorts, the long leg of an anomaly is the value-weighted portfolio of stocks in the lowest decile of the anomaly measure, and the short leg contains the stocks in the highest decile, with a high value of the measure being associated with lower return. In Panel B, long/short legs are neutralized with respect to size. That is, we first form size deciles by sorting on the previous month's market value. Within each size decile, we then create ten deciles formed by sorting on the anomaly variable. Finally, we form the anomaly's decile portfolios, with each portfolio pooling the stocks in a given anomaly decile across the size groups, again with value weighting. Panel B omits the size anomaly, whose alpha equals zero by construction with size-neutral sorts. Our sample period is January 2000 through December 2016 (204 months). All t -statistics are based on the heteroskedasticity-consistent standard errors of [White \(1980\)](#).

Category	Anomaly	$\bar{R}$	α	β	$t(\bar{R})$	$t(\alpha)$	$t(\beta)$
Panel A: Unconditional sorts							
Size	Market cap	1.09	0.97	0.18	1.92	1.81	1.90
Value	EP	1.27	1.37	-0.16	2.58	2.93	-2.15
Value	BM	1.14	1.14	0.01	2.08	2.13	0.06
Value	CP	0.73	0.70	0.04	1.67	1.69	0.55
Profitability	ROE	0.83	0.93	-0.15	1.77	2.11	-2.09
Volatility	1-Month vol.	0.81	1.03	-0.34	1.64	2.31	-5.55
Volatility	MAX	0.57	0.81	-0.36	1.26	2.02	-6.39
Reversal	1-Month return	1.47	1.49	-0.02	2.96	3.06	-0.22
Turnover	12-Month turn.	0.33	0.53	-0.29	0.63	1.09	-3.46
Turnover	1-Mo. abn. turn.	1.14	1.27	-0.19	2.44	2.92	-2.66
Panel B: Size-neutral sorts							
Value	EP	1.80	1.89	-0.14	4.32	4.72	-2.25
Value	BM	1.14	1.10	0.05	2.23	2.22	0.64
Value	CP	0.78	0.76	0.04	2.27	2.25	0.71
Profitability	ROE	1.45	1.50	-0.07	3.90	4.11	-1.30
Volatility	1-Month Vol.	0.66	0.90	-0.37	1.41	2.19	-6.20
Volatility	MAX	0.39	0.60	-0.32	0.93	1.61	-6.14
Reversal	1-Month return	1.67	1.65	0.02	3.65	3.68	0.32
Reversal	12-Month turn.	0.51	0.74	-0.34	1.06	1.74	-4.94
Turnover	1-Mo. abn. turn.	1.29	1.39	-0.15	3.19	3.68	-2.56

Table 6: CAPM alphas for anomalies

Table 8

FF-3 alphas and factor loadings for anomalies.

For each of ten anomalies, the table reports the monthly long-short return spread's CH-3 alpha and factor loadings. For each anomaly, the regression estimated is

$$R_t = \alpha + \beta_{MKT} MKT_t + \beta_{SMB} FFSMB_t + \beta_{FFHML} FFHML_t + \epsilon_t,$$

where R_t is the anomaly's long-short return spread in month t , MKT_t is the excess market return, SMB_t is FF-3's size factor, and $FFHML_t$ is the BM-based value factor. In Panel A, for the unconditional sorts, the long leg of an anomaly is the value-weighted portfolio of stocks in the lowest decile of the anomaly measure, and the short leg contains the stocks in the highest decile, with a high value of the measure being associated with lower return. In Panel B, long/short legs are neutralized with respect to size. That is, we first form size deciles by sorting on the previous month's market value. Within each size decile, we then create ten deciles formed by sorting on the anomaly variable. Finally, we form the anomaly's decile portfolios, with each portfolio pooling the stocks in a given anomaly decile across the size groups, again with value weighting. Panel B omits the size anomaly, whose alpha equals zero by construction with size-neutral sorts. Our sample period is January 2000 through December 2016. All t -statistics are based on the heteroskedasticity-consistent standard errors of [White \(1980\)](#).

Category	Anomaly	α	β_{MKT}	β_{SMB}	β_{FFHML}	$t(\alpha)$	$t(\beta_{MKT})$	$t(\beta_{SMB})$	$t(\beta_{FFHML})$
Panel A: Unconditional sorts									
Size	Market cap	0.16	0.04	1.55	-0.11	1.36	1.86	34.29	-2.43
Value	EP	1.54	-0.07	-0.98	0.48	5.57	-1.63	-13.29	6.54
Value	BM	-0.28	0.00	0.12	1.60	-1.25	0.09	1.48	30.61
Value	CP	0.63	0.08	-0.49	0.43	1.40	1.31	-3.84	2.31
Profitability	ROE	1.75	-0.06	-1.01	-0.28	5.67	-1.36	-12.74	-3.08
Volatility	1-Month vol.	0.83	-0.30	-0.40	0.52	2.11	-5.41	-3.07	4.18
Volatility	MAX	0.74	-0.33	-0.28	0.28	1.85	-5.70	-1.79	1.96
Reversal	1-Month return	0.94	-0.06	0.53	0.28	1.97	-0.83	3.50	1.58
Turnover	12-Month turn.	0.83	-0.19	-1.08	0.38	2.96	-4.97	-13.10	3.72
Turnover	1-Mo. abn. turn.	1.34	-0.21	0.16	-0.20	2.86	-2.76	0.95	-0.99
Panel B: Size-neutral sorts									
Value	EP	1.76	-0.09	-0.53	0.52	5.49	-1.79	-6.52	6.17
Value	BM	-0.01	0.07	-0.10	1.39	-0.04	1.75	-1.36	20.12
Value	CP	0.52	0.06	-0.23	0.44	1.73	1.33	-2.23	4.05
Profitability	ROE	2.01	-0.04	-0.36	-0.35	5.72	-0.71	-3.75	-3.38
Volatility	1-Month vol.	0.76	-0.33	-0.33	0.40	2.06	-6.04	-2.64	3.47
Volatility	MAX	0.43	-0.30	-0.16	0.31	1.14	-5.67	-1.10	2.60
Reversal	1-Month return	1.21	-0.01	0.35	0.29	2.55	-0.09	2.63	1.78
Turnover	12-Month turn.	0.80	-0.28	-0.68	0.39	2.46	-5.89	-6.37	3.74
Turnover	1-Mo. abn. turn.	1.37	-0.17	0.21	-0.12	3.26	-2.82	1.43	-0.68

Table 8: FF-3 alphas and loadings for anomalies

6.6 Table 9 and Table 10: model comparison and CH-4

Read:

- Table 9 summarizes how well each model explains all anomalies.

- CH-3 has the smallest average absolute alpha and the most favorable GRS performance among CAPM, FF-3, and CH-3.
- Table 10 shows that adding the turnover-based PMO factor produces CH-4.
- CH-4 further reduces alphas for reversal and turnover-related anomalies.

Economic message: CH-3 is the main benchmark model; CH-4 is a targeted extension that accounts for sentiment-driven turnover patterns in China.

12-Month turn.	0.80	-0.28	-0.68	0.39	2.46	-5.89	-6.37
1-Mo. abn. turn.	1.37	-0.17	0.21	-0.12	3.26	-2.82	1.43

Table 9
Comparing the abilities of models to explain anomalies.
The table reports measures summarizing the degree to which anomalies produce alphas under three different factor models: CAPM, FF-3, and CH-3. Also reported are measures for “unadjusted” return spreads (i.e. for a model with no factors). For each model, the table reports the average absolute monthly alpha (in percent), average absolute t -statistic, the Gibbons-Ross-Shanken (1989) “GRS” F -statistic with associated p -value, and the number of anomalies for which the model produces the smallest absolute alpha among the four models. In Panel A, for the unconditional sorts, the long leg of an anomaly is the value-weighted portfolio of stocks in the lowest decile of the anomaly measure, and the short leg contains the stocks in the highest decile, with a high value of the measure being associated with lower return. In Panel B, long/short legs are neutralized with respect to size. That is, we first form size deciles by sorting on the previous month’s market value. Within each size decile, we then create ten deciles formed by sorting on the anomaly variable. Finally, we form the anomaly’s decile portfolios, with each portfolio pooling the stocks in a given anomaly decile across the size groups, again with value weighting. Two versions of the GRS test are reported. In Panel A, GRS_{10} uses all ten anomalies, while GRS_5 excludes the anomalies for size, BM, and EP, which are variables used to construct factors. Panel B omits the size anomaly, whose alpha equals zero by construction with size-neutral sorts. All t -statistics are based on the heteroskedasticity-consistent standard errors of White (1980). The sample period is from January 2000 through December 2016 (204 months).

Measure	Unadjusted	CAPM	FF-3	CH-3
Panel A: Unconditional sorts				
Average $ \alpha $	0.94	1.02	0.90	0.45
Average $ t $	1.89	2.21	2.70	1.12
GRS_{10}	7.30	7.31	6.00	1.49
p_{10}	< 0.0001	< 0.0001	< 0.0001	0.15
GRS_5	4.40	4.45	6.86	1.74
p_5	0.0002	0.0001	0.0001	0.10
Panel B: Size-neutral sorts				
Average $ \alpha $	1.08	1.17	0.99	0.47
Average $ t $	2.55	2.72	1.97	1.07
GRS_{10}	8.24	8.08	7.97	1.97
p_{10}	< 0.0001	< 0.0001	< 0.0001	0.05
GRS_5	8.15	8.10	9.11	2.33
p_5	< 0.0001	< 0.0001	< 0.0001	0.03

Table 9: comparing anomaly-pricing ability

Table 10
Anomaly alphas under a four-factor model.
For each of ten anomalies, the table reports the monthly long-short return spread’s CH-4 alpha and factor loadings. For each anomaly, the regression estimated is

$$R_t = \alpha + \beta_{MKT} MKT_t + \beta_{SMB} SMB_t + \beta_{VMG} VMG_t + \beta_{PMO} PMO_t + \epsilon_t,$$

in which R_t is the anomaly’s long-short return spread in month t , MKT_t is the excess market return, SMB_t is CH-3’s size factor, VMG_t is the EP-based value factor, and PMO_t (pessimistic minus optimistic) is the sentiment factor based on abnormal turnover. In Panel A, for the unconditional sorts, the long leg of an anomaly is the value-weighted portfolio of stocks in the lowest decile of the anomaly measure, and the short leg contains the stocks in the highest decile, with a high value of the measure being associated with lower return. In Panel B, long/short legs are neutralized with respect to size. That is, we first form size deciles by sorting on the previous month’s market value. Within each size decile, we then create ten deciles formed by sorting on the anomaly variable. Finally, we form the anomaly’s decile portfolios, with each portfolio pooling the stocks in a given anomaly decile across the size groups, again with value weighting. Panel B omits the size anomaly, whose alpha equals zero by construction with size-neutral sorts. Our sample period is January 2000 through December 2016. All t -statistics are based on the heteroskedasticity-consistent standard errors of White (1980).

Category	Anomaly	α	β_{MKT}	β_{SMB}	β_{VMG}	β_{PMO}	$t(\alpha)$	$t(\beta_{MKT})$	$t(\beta_{SMB})$	$t(\beta_{VMG})$	$t(\beta_{PMO})$
Panel A: Unconditional sorts											
Size	Market cap	0.23	0.05	1.50	-0.42	0.01	1.41	3.05	39.38	-6.88	0.17
Value	EP	0.02	0.00	-0.46	1.34	0.04	0.08	0.07	-5.38	13.15	0.33
Value	BM	0.75	0.03	-0.02	0.43	-0.13	1.04	0.32	-0.09	1.55	-0.45
Value	CP	0.31	0.09	-0.26	0.65	-0.20	0.57	1.43	-1.81	3.60	-1.10
Profitability	ROE	-0.29	-0.03	-0.36	1.28	-0.10	-0.68	-0.70	-3.21	8.74	-0.96
Volatility	1-Month vol.	-0.27	-0.16	-0.27	0.59	0.72	-0.51	-2.71	-1.85	3.27	5.13
Volatility	MAX	-0.59	-0.18	-0.13	0.44	0.88	-1.64	-3.07	-0.90	2.91	7.91
Reversal	1-Month return	0.49	0.02	0.54	0.04	0.46	0.87	0.29	3.19	0.18	2.48
Turnover	12-Month turn.	0.04	-0.11	-0.94	0.64	0.43	0.11	-3.36	-10.60	5.09	3.69
Turnover	1-Mo. abn. turn.	-0.00	-0.01	0.07	-0.27	1.44	-0.01	-0.32	0.68	-2.43	16.47
Panel B: Size-neutral sorts											
Value	EP	0.43	-0.04	-0.03	1.28	-0.12	1.42	-0.79	-0.35	11.74	-1.09
Value	BM	0.57	0.12	-0.18	0.46	0.09	0.82	1.58	-1.04	1.83	0.39
Value	CP	0.19	0.08	-0.03	0.57	-0.12	0.49	1.56	-0.26	3.67	-0.92
Profitability	ROE	-0.30	0.02	0.35	1.23	-0.05	-0.76	0.51	3.67	8.99	-0.40
Volatility	1-Month vol.	-0.27	-0.21	-0.20	0.51	0.63	-0.59	-3.95	-1.37	2.87	4.90
Volatility	MAX	-0.77	-0.17	0.00	0.45	0.74	-2.05	-2.86	0.01	2.95	5.81
Reversal	1-Month return	0.71	0.07	0.40	0.12	0.42	1.28	1.14	2.62	0.88	2.60
Turnover	12-Month turn.	-0.07	-0.19	-0.53	0.61	0.44	-0.19	-4.16	-4.38	3.77	3.39
Turnover	1-Mo. abn. turn.	0.17	-0.00	0.17	-0.17	1.21	0.67	-0.05	1.83	-1.90	15.23

Table 10: CH-4 anomaly alphas and loadings

6.7 Appendix evidence: anomaly literature and alternative factor models

Read:

- Table A1 compiles Chinese anomalies in the literature and reports CAPM alphas for the authors’ sample.
- Table A5 shows that CH-3 also dominates a China replication of the Fama–French five-factor model.
- Figure A1 shows Bayesian posterior model probabilities: CH-3 dominates FF-3 across a wide range of priors.

Economic message: The paper’s conclusions are robust to alternative comparisons and are not only a frequentist GRS-test result.

Table A1

Chinese anomalies in the literature.

The table provides a compilation of the anomalies reported by one or more studies as being significant in China. The studies analyzing each anomaly are listed, including those reporting an anomaly to be insignificant (identified as such in the table). For each anomaly variable, we also report CAPM alphas for both unconditional and conditional sorts. For the unconditional sorts, the long leg of an anomaly is the value-weighted portfolio of stocks in the lowest decile of the anomaly measure, and the short leg contains the stocks in the highest decile, with a high value of the measure being associated with lower return. For the conditional sorts, long/short legs are neutralized with respect to size. That is, we first form size deciles by sorting on the previous month's market value. Within each size decile, we then create ten deciles formed by sorting on the anomaly variable. Finally, we form the anomaly's decile portfolios, with each portfolio pooling the stocks in a given anomaly decile across the size groups, again with value weighting. Our sample period is January 2000 through December 2016 (204 months). All *t*-statistics (in parentheses) are based on the heteroskedasticity-consistent standard errors of White (1980).

Category	Anomaly	References	CAPM alpha (monthly %)	
			Unconditional	Size-neutral
Size	Market cap	Wang and Xu (2004), Eun and Huang (2007), Cheung et al. (2015), Cakici et al. (2017), Hsu et al. (2018), Carpenter et al. (2018), and Hu et al. (2019). Reported insignificant: Chen et al. (2010) and Cheung et al. (2015).	0.97 (1.81)	–
Value	EP	Cakici et al. (2017) and Hsu et al. (2018). Reported insignificant: Chen et al. (2010) and Hu et al. (2019).	1.37 (2.93)	1.89 (4.72)
Value	BM	Wang and Xu (2004), Eun and Huang (2007), Chen et al. (2010), Cheung et al. (2015), Cakici et al. (2017), Hsu et al. (2018), and Carpenter et al. (2018). Reported insignificant: Hu et al. (2019).	1.14 (2.13)	1.10 (2.22)
Value	CP	Cakici et al. (2017). Reported insignificant: Wang and Di Iorio (2007) and Chen et al. (2010).	0.70 (1.69)	0.76 (2.25)
Profitability	ROE	Guo et al. (2017). Reported insignificant: Li et al. (2007).	0.93 (2.11)	1.50 (4.11)
Volatility	1-Mo. vol.	Cheung et al. (2015), Cakici et al. (2017), and Hsu et al. (2018). Reported insignificant: Chen et al. (2010).	1.03 (2.31)	0.90 (2.19)
Volatility	MAX	Carpenter et al. (2018).	0.81 (2.02)	0.60 (1.61)
Reversal	1-Month return	Cakici et al. (2017), Hsu et al. (2018), and Carpenter et al. (2018). Reported insignificant: Cheung et al. (2015).	1.49 (3.06)	1.65 (3.68)
Turnover	12-Month turn.	Zhang and Liu (2006) and Eun and Huang (2007). Reported insignificant: Chen et al. (2010).	0.53 (1.09)	0.74 (1.74)
Turnover	1-Mo. abn. turn.	Li and Wu (2003) and Zhang and Liu (2006).	1.27 (2.92)	1.39 (3.68)
Investment	Asset growth	Chen et al. (2010). Reported insignificant: Hsu et al. (2018), Guo et al. (2017), and Lin (2017).	0.22 (0.72)	–0.05 (–0.20)
Accruals	Accruals	Li et al. (2011) and Hsu et al. (2018). Reported insignificant: Chen et al. (2010).	0.08 (0.39)	–0.15 (–0.70)
Accruals	NOA	Chen et al. (2010) and Hsu et al. (2018).	0.38 (1.03)	0.42 (1.22)
Illiquidity	Amihud illiq.	Carpenter et al. (2018) and Chen et al. (2010).	0.83 (1.62)	0.63 (1.55)

*Table A1: Chinese anomalies in the literature***Table A5**

Abilities of models CH-3 and FF-5 to explain each other's factors.

Panel A reports a factor's estimated monthly alpha (in percent) with respect to the other model (with White, 1980, heteroskedasticity-consistent *t*-statistics in parentheses). Panel B computes the Gibbons et al. (1989) *F*-test of whether a given model produces zero alphas for the factors of the other model (*p*-value in parentheses). The sample period is January 2000 through December 2016.

Factors	Alphas with respect to:	
	CH-3	FF-5
<i>Panel A: Alpha (t-statistics)</i>		
SMB	–	0.14 (2.41)
VMG	–	0.43 (4.39)
FFSMB	0.01 (0.18)	–
FFHML	0.34 (0.96)	–
FFRMW	–0.10 (–0.86)	–
FFCMA	–0.08 (–0.51)	–
<i>Panel B: GRS F-statistics (p-value)</i>		
SMB, VMG	–	8.43 (0.0003)
FFSMB, FFHML, FFRMW, FFCMA	0.29 (0.88)	–

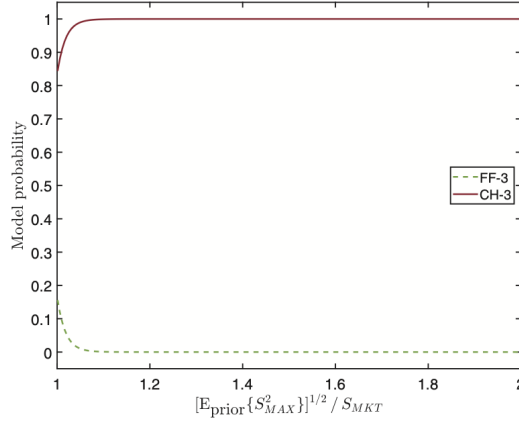
Table A5: CH-3 versus FF-5

Fig. A1. Model probabilities comparing model CH-3 to model FF-3. The figure displays Bayesian posterior model probabilities for the two-way model comparison. The value on the horizontal axis is the square root of the prior expected maximum Sharpe ratio achievable by combining the model's factors, divided by the observed Sharpe ratio of the market. Prior model probabilities are equal. The sample period is from January 2000 through December 2016 (204 months).

7 Short summary

- The paper builds a China-specific factor model rather than mechanically importing the Fama–French U.S. procedure.
- The first key adjustment is to exclude the smallest 30% of listed firms, because these stocks are strongly affected by shell value in reverse mergers.
- The second key adjustment is to use EP rather than BM to form the value factor.
- The resulting CH-3 model has a market factor, a shell-free size factor, and an EP-based value factor.
- Size and value premiums are both large and significant in China, each exceeding 1% per month over 2000–2016.
- CH-3 prices the FF-style China factors, while the FF-style model fails to price CH-3 factors.

- CH-3 explains most documented Chinese anomalies, including profitability and volatility anomalies.
- CH-3 leaves some turnover and reversal patterns, which motivate CH-4 with a turnover-based sentiment factor.
- The paper’s main lesson is that Chinese asset-pricing factors must reflect Chinese institutions, especially IPO constraints and reverse-merger shell value.

8 Conclusion

8.1 Core conclusion

Liu, Stambaugh, and Yuan show that size and value are powerful characteristics in China, but the right way to measure them differs from the U.S. The smallest stocks should be excluded from factor construction because their prices contain substantial potential shell value. The value factor should be based on earnings-price rather than book-to-market.

8.2 Economic intuition

China’s IPO approval regime creates scarcity value for listed shells. That scarcity value is concentrated among very small listed firms, so small-stock returns do not purely reflect small operating businesses. At the same time, earnings yield is more informative than book-to-market about expected returns in China. Once these two facts are incorporated, a three-factor model explains many anomalies that a standard FF-style model misses.

8.3 Takeaway

The paper’s broad message is that factor models should be institutionally grounded. A factor model that works in one market can fail in another if its characteristics are contaminated by local institutions or if different accounting variables better capture expected-return variation. For China, the key ingredients are a shell-free size factor and an EP-based value factor.